

Figure 1. PRISMA 2020 flow - two-arm identification (OpenAlex database + web/citation)

Identification

Screening

Included

Arm A - bibliographic database (OpenAlex API)

Arm B - web deep-search + citation chasing

Records retrieved from OpenAlex
(30 topic queries × 50)
n = 1500

Unique records after de-duplication
n = 957
(duplicates removed n = 543)

Records topicality-screened
on-topic n = 165

Off-topic / generic LLM works
excluded at screening
n = 792

Records identified via web search
(59 queries) n = 472
+ citation chasing n = 28

Unique records after de-duplication
n = 304
(duplicates removed n = 196)

Records screened
screened-in n = 129

Excluded at screening
(off-topic / no comparison / type)
n = 175

Reports assessed for eligibility (merged, cross-source duplicates removed) n = 263

Excluded, with reasons
no direct human comparison n = 22
LLM judge used as tool only n = 11
method only, no human eval n = 8
grey / non-peer literature n = 5

Studies included in the review
(quantitative + qualitative synthesis)
n = 217